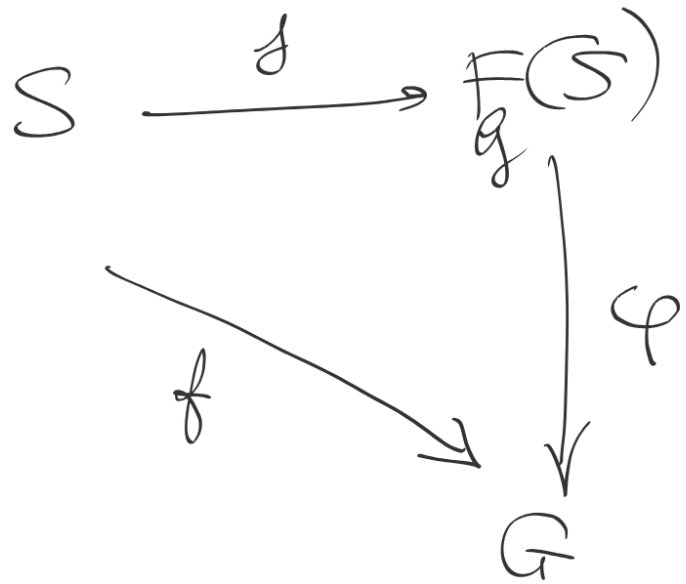
Algebra Monsoon 2026 Lecture 6

Free groups

Let S be a set.

Problem Does there exist a pair $(F(S), j)$ consisting of a group $F(S)$ and a mapping $j : S \to F(S)$ such that for every pair (G, f) consisting of a group G and a mapping $f : S \to G$, there is a unique morphism of groups $\varphi : F(S) \to G$ such that $f = \varphi \circ j$?

(The triangle



should commute.)

Such a pair is called a free group on the set S.

If the free group $(F_g(S), j)$ exists, it unique up to a unique isomorphism:

Proposition If (F_1, j_1) and (F_2, j_2) are free groups on S, then the unique morphism of groups $F_1 \xrightarrow{\varphi} F_2$ such that $j_2 = \varphi \circ j_1$ is an isomorphism. □

Construction of the free group

Consider the direct sum

$$S^{\pm} = S \amalg S$$

of two copies of S
along with the two injections

$$S \underset{s \mapsto s^-}{\overset{s \mapsto s^+}{\rightrightarrows}} S^{\pm}$$

for $s \in S$. The canonical involution of the set $S^{\pm}$ is

$$\left.\begin{aligned} s^+ &\longmapsto s^- \\ s^- &\longmapsto s^+ \end{aligned}\right\} \text{ for every } s \in S.$$

Let $F_m(S^{\pm})$ be the free monoid on the set $S^{\pm}$.

(For any set T, the mapping $T \to F_m(T)$ is injective, for if two distinct $t_1, t_2 \in T$ had the same image in $F_m(T)$, a mapping from T into a magma sending t_1, t_2 to two distinct elements will not factor through $L(T)$.)

We consider $S^{\pm}$ as a subset of $F_m(S^{\pm})$

Let C be the smallest congruence on the monoid $F_m(S^{\pm})$ such that

$$s^{+}s^{-} \underset{C}{\sim} e \quad , \quad s^{-}s^{+} \underset{C}{\sim} e$$

for every $s \in S$.

Consider the quotient monoid

$$\pi : F_m(S^{\pm}) \longrightarrow F_g(S) = F_m(S)/C$$

Proposition The monoid $F_g(S)$ is a group.

Proof For $s \in S$, we have

$$\pi(s^+)\pi(s^-) = \pi(s^+ s^-)$$
$$= e$$

(because $s^+ s^- \underset{C}{\sim} e$). Similarly,

$$\pi(s^-)\pi(s^+) = e.$$

Therefore $\pi(s^+), \pi(s^-) \in F(S)^\times$ for every $s \in S$, and in fact

$$\pi(s^+)^{-1} = \pi(s^-)$$
$$\pi(s^-)^{-1} = \pi(s^+)$$

As the monoid $F_g(S)$ is generated by $\pi_*(S^{\pm})$ (because π is surjective and the monoid $F_m(S^{\pm})$ is generated by $S^{\pm}$), every element of $F_g(S)$ is invertible. Hence $F_g(S)$ is a group.

□

Define $j_+ : S \to F_g(S)$ by

$$j_+(s) = \pi(s^+)$$

for every $s \in S$. (We could have equally chosen $j_-(s) = \pi(s^-)$).

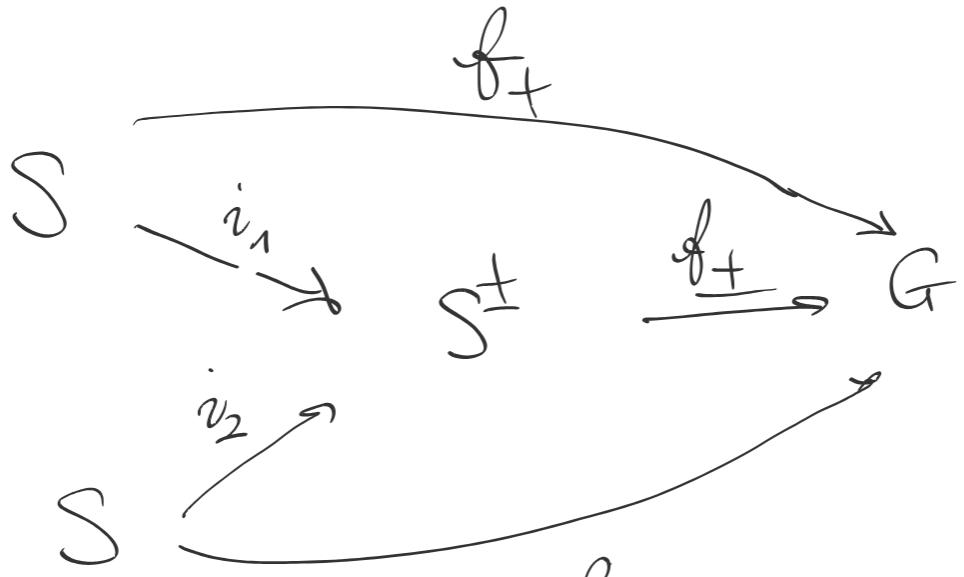
Let us prove that $(F_g(S), j_+)$ is the free group on S:

Proposition For every mapping $f_+ : S \to G$ into a group, there is a unique morphism of groups $\varphi : F_g(S) \to G$ such that $f_+ = \varphi \circ j_+$.

Proof Let

$$f_\pm : S^\pm \longrightarrow G$$

be the unique mapping such that



commutes, where $f_- = f_+(s)^{-1}$ for every $s \in S$. Note that we have

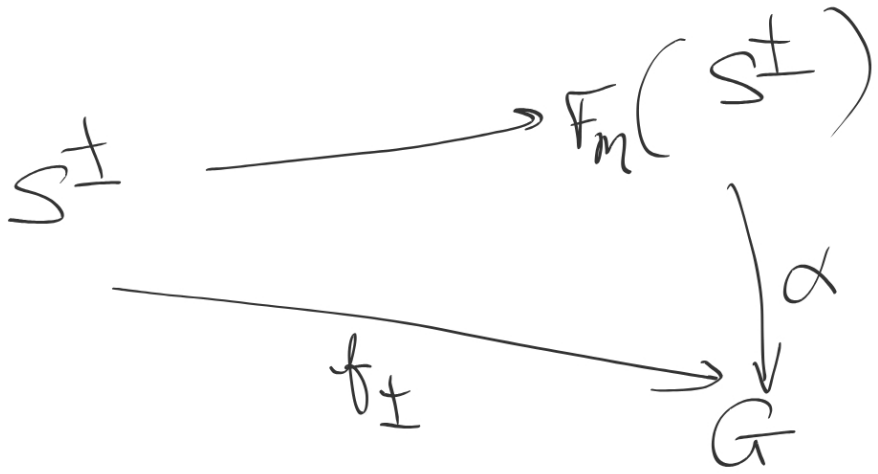
$$f_\pm(s^+) f_\pm(s^-) = f_+(s) f_-(s) = e$$

and $f_\pm(s^-) f_\pm(s^+) = f_-(s) f_+(s) = e$.

By the universal property of the free monoid $F_m(S^{\pm})$, there is a unique morphism of monoids

$$\alpha : F_m(S^{\pm}) \longrightarrow G$$

making the triangle



commute.

For every $s \in S$, we have

$$\alpha(s^+ s^-) = \alpha(s^+)\,\alpha(s^-)$$
$$= f_+(s^+) f_+(s^-)$$
$$= e$$

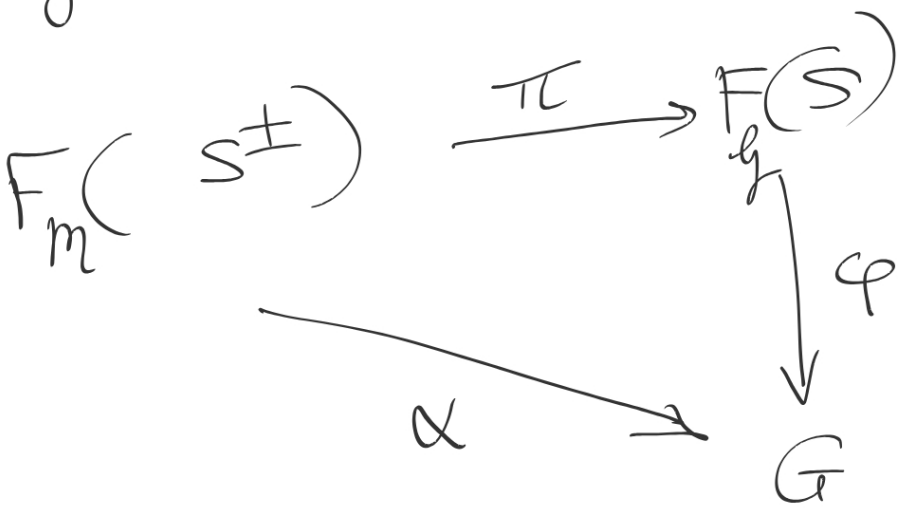
and similarly

$$\alpha(s^- s^+) = e.$$

Therefore, by the universal property of the quotient $\pi : F_m(S^{\pm}) \to F_g(S)$, there is a unique morphism of monoids

$$\varphi : F_g(S) \to G$$

making the triangle



Commute. We have, for $s \in S$,

$$\varphi(j_+(s)) = \varphi(\pi(s^+))$$

$$= \alpha(s^+)$$

$$= f_\pm(s^+)$$

$$= f_+(s),$$

so that $f_+ = \varphi \circ j_+$.

Note that φ is a morphism of <u>groups</u> because it is a morphism of magmas and the source and the target are groups. □

Reformulation

Proposition Let S be a set and $(F_{Gp}(S), j_+)$ the free group on S.

For every group G, the mapping

$$\mathrm{Hom}_{Gp}(F_{Gp}(S), G) \longrightarrow \mathrm{Hom}_{Ens}(S, G)$$

($\leftarrow$ groups, sets $\rightarrow$)

$$\varphi \longmapsto \varphi \circ j_+$$

is a bijection.

... mapping and

is a bijection.

Proposition Let $\alpha : T \to S$ be a mapping and $\beta : G \to H$ a morphism of groups.

The diagram

$$\begin{array}{ccc}
\mathrm{Hom}_g(F_g(S), G) & \xrightarrow{\sim} & \mathrm{Hom}_{\mathcal{E}}(S, G) \\
\downarrow \circ F_g(\alpha) & & \downarrow \circ \alpha \\
\mathrm{Hom}_g(F_g(T), G) & \xrightarrow{\sim} & \mathrm{Hom}_{\mathcal{E}}(T, G) \\
\downarrow \beta \circ & & \downarrow \beta \circ \\
\mathrm{Hom}_g(F_g(T), H) & \xrightarrow{\sim} & \mathrm{Hom}_{\mathcal{E}}(T, H)
\end{array}$$

commutes, $F_g(\alpha) : F_g(T) \to F_g(S)$ is the unique morphism of groups extending the mapping $T \to S \to F_g(S)$.

Free modules

Let S be a set and let $\mathbb{Z}^{(S)}$ be the set of all mappings

$$x : S \longrightarrow \mathbb{Z}$$

such that $x(s) = 0$ for all but finitely many $s \in S$.

For $x, y \in S$, define

$$x + y \in S$$

by $(x+y)(s) = x(s) + y(s)$ for all $s \in S$. This law of composition makes $\mathbb{Z}^{(S)}$ into a module, with the constant mapping 0 as the neutral element.

For $s \in S$, let $e_s \in \mathbb{Z}^{(S)}$ be the mapping such that

$$e_s(s) = 1 \quad \text{and} \quad e_s(t) = 0$$

for $t \neq s$. For every $x \in \mathbb{Z}^{(S)}$, we have

$$x = \sum_{s \in S} x(s) e_s ,$$

so that if

$$\sum_{s \in S} x(s) e_s = \sum_{s \in S} y(s) e_s$$

for some $x, y \in \mathbb{Z}^{(S)}$, then

$$x = y .$$

Define $i : S \to \mathbb{Z}^{(S)}$ by $i(s) = e_s$. We have seen that the module $\mathbb{Z}^{(S)}$ is generated by the subset $i_*(S)$

We claim that $(\mathbb{Z}^{(S)}, i)$ is the free module on the set S:

Proposition For every module A, the mapping

$$\begin{aligned} \mathrm{Hom}_{\mathcal{A}}(F(S), A) &\longrightarrow \mathrm{Hom}_{\mathcal{E}}(S, A) \\ \varphi &\longmapsto \varphi \circ i \end{aligned}$$

($\mathcal{A}$: modules; $\mathcal{E}$: sets)

is a bijection.

Exercise

Let S be a set and $(F_g(S), j)$ the free group on S. Let C be the smallest congruence on $F_g(S)$ such that $xy \sim_C yx$ for all $x, y \in F_g(S)$.

Show that $F_g(S)/C$ is a free module on S.

Make the resulting isomorphism

$$F_g(S)/C \longrightarrow \mathbb{Z}^{(S)}$$

explicit by specifying the composite morphism of monoids

$$F_m(S) \longrightarrow \mathbb{Z}^{(S)}$$

(where $F_m(S)$ is the free monoid on S).

Groups defined by generators and relators

Let S be a set and $R \subset F_{fg}(S)$ a subset of the free group on S.

Let C be the smallest congruence on $F_{fg}(S)$ such that $r \sim_C e$ for every $r \in R$. The quotient group

$$\langle S \mid R \rangle = F_{fg}(S)/C$$

is said to be the group defined by the set of generators S and the set of relators R.

Denote by j the composite map

$$S \longrightarrow F_{fg}(S) \xrightarrow{\pi} \langle S \mid R \rangle ;$$

the group $\langle S \mid R \rangle$ is generated by the subset $j_*(S)$.

Proposition (Universal property of the group defined by generators and relators)

Let $f : S \to A$ be a mapping into a group and $\varphi : F_{fg}(S) \to A$ the unique morphism extending f. If $\varphi(r) = e$ for every $r \in R$, there exists a unique morphism of groups $\langle S \mid R \rangle \xrightarrow{\overline{\varphi}} G$ making the triangle

$$\begin{array}{ccc} F_{fg}(S) & \xrightarrow{\pi} & \langle S \mid R \rangle \\ & \searrow^{\varphi} & \downarrow{\scriptstyle \overline{\varphi}} \\ & & A \end{array}$$

Commute ($\varphi = \overline{\varphi} \circ \pi$). □

Coproduct of a family of groups

The product of family of groups $(G_\lambda)_{\lambda \in \Lambda}$ is a group $\Pi = \prod_{\lambda \in \Lambda} G_\lambda$ together with a family of morphisms

$$\pi_\lambda : \Pi \longrightarrow G_\lambda \qquad (\lambda \in \Lambda)$$

such that for every group H, the mapping

$$\mathrm{Hom}_{\mathcal{G}}(H, \Pi) \longrightarrow \prod_{\lambda \in \Lambda} \mathrm{Hom}_{\mathcal{E}}(H, G_\lambda)$$

($\mathcal{G}$: groups; $\mathcal{E}$: sets)

$$\varphi \longmapsto (\pi_\lambda \circ \varphi)_{\lambda \in \Lambda}$$

is a bijection.

The coproduct of the family $(G_\lambda)_\lambda$ is a group $\Sigma = \coprod_{\lambda \in \Lambda} G_\lambda$ together with a family of morphisms

$$\iota_\lambda : G_\lambda \longrightarrow \Sigma$$

such that for every group H, the mapping

$$\mathrm{Hom}_g(\Sigma, H) \to \prod_{\lambda \in \Lambda} \mathrm{Hom}_\varepsilon(H, G_\lambda)$$

$$\varphi \longmapsto (\varphi \circ \iota_\lambda)_{\lambda \in \Lambda}$$

is a bijection.

Does the coproduct exist?

We know that if it exists, it is unique up to a unique isomorphism.

Construction of Σ:

Take $S = \coprod_{\lambda \in \Lambda} G_\lambda$, so that elements of S are of the form (λ, a) where $\lambda \in \Lambda$ and $a \in G_\lambda$.

Take R be the set of relators

$$(\lambda, a)(\lambda, b)(\lambda, ab)^{-1}$$

(where $\lambda \in \Lambda$ and $a, b \in G_\lambda$)

and $$(\lambda, e)(\mu, e)^{-1}$$

(where $\lambda, \mu \in \Lambda$ and the e's are the neutral elements of G_λ, G_μ respectively)

Put $\Sigma = \langle S | R \rangle$,

the group defined by the set of generators S and the set of relators R.

For $\lambda \in \Lambda$ and $a \in G_\lambda$, let

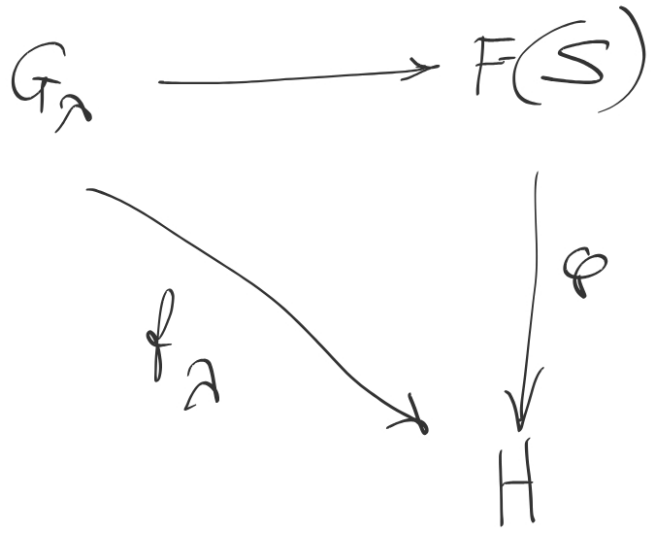
$$\iota_\lambda(a) \in \Sigma$$

be the image of (λ, a).

The mapping $\iota_\lambda : G_\lambda \to \Sigma$ is a monomorphism.

Proposition $\left(\Sigma, \left(\iota_\lambda : G_\lambda \to \Sigma\right)_{\lambda \in \Lambda}\right)$ is the coproduct of the family of groups $(G_\lambda)_{\lambda \in \Lambda}$.

Proof Let $f_\lambda : G_\lambda \to H$ be a family of morphism. By the universal property of the free group, there is a unique morphism φ making the triangle



$$\varphi\big((\lambda,a)(\lambda,b)(\lambda,ab)^{-1}\big)$$
$$= f_\lambda(a)\, f_\lambda(b)\, f_\lambda(ab)^{-1}$$
$$= e$$

for every $\lambda \in \Lambda$ and $a, b \in G_\lambda$,

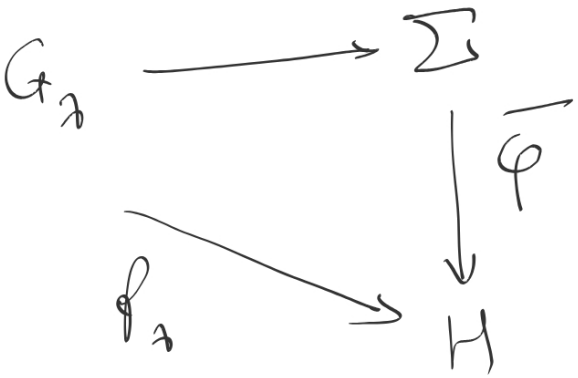
and

$$\varphi\big((\lambda,e)(\mu,e)^{-1}\big)$$
$$= f_\lambda(e)\, f_\mu(e)^{-1}$$
$$= e\, e^{-1} = e$$

for every $\lambda, \mu \in \Lambda$

the quotient now says that there is a unique morphism $\bar{\varphi}$

making the triangle



commute for every $\lambda \in \Lambda$. □

2 The underlying set of $\coprod_{\lambda \in \Lambda}^{\mathcal{G}} G_\lambda$ is $\coprod_{\lambda \in \Lambda}^{\mathcal{E}_0} G_\lambda$

but the underlying set of $\coprod_{\lambda \in \Lambda}^{\mathcal{G}} G_\lambda$ is <u>not</u> $\coprod_{\lambda \in \Lambda}^{\mathcal{E}_0} G_\lambda$.

For example

$C_2 \coprod_{\mathcal{G}} C_3$ is infinite

whereas $C_2 \coprod_{\mathcal{E}_0} C_3$ has only

five elements

For a set S, $(G_s)_{s \in S}$ be

the family of groups such that

$$G_s = \mathbb{Z} \text{ for every } s \in S.$$

We have the family of morphisms

$$\iota_s : G_s \longrightarrow \coprod_{s \in S} G_s ;$$

Consider the mapping

$$f : S \longrightarrow \coprod_{s \in S} G_s$$

defined by $f(s) = \iota_s(1)$

for every $s \in S$.

Λ of the

for any $s \in S$.

By the universal property of the free group $F_g(S)$, there is a unique morphism

$$\varphi : F_g(S) \longrightarrow \coprod_{s \in S} G_s$$

such that

$$\varphi(s) = i_s(1)$$

for every $s \in S$.

Proposition φ is an isomorphism of groups.

Proof Consider the family of morphisms

$$g_s : G_s \longrightarrow F_G(S)$$

such that $g_s(1) = s$.

By the universal property of the coproduct, there is a unique morphism

$$\coprod_{s \in S} G_s \xrightarrow{\psi} F_G(S)$$

such that $g_s = \psi \circ i_s$

for every $s \in S$.

For every $s \in S$, we have

$$\psi(\varphi(s)) = \psi(i_s(1))$$

$$= g_s(1)$$

$$= s,$$

and, as S generates $F_G(S)$ as a group, we have

$$\psi \circ \varphi = Id_{F_G(S)}.$$

Next, we know that $\coprod_{s \in S}^{gp} G_s$

is generated by $\bigcup_{s \in S} i_{s*}(G_s)$

and each $i_{s*}(G_s)$ is generated

by $i_s(1)$, so it enough to

check that

$$\varphi(\psi(i_s(1))) = i_s(1)$$

for every $s \in S$ to conclude

that $\varphi \circ \psi = Id_{\coprod_{s \in S}^{gp} G_s}$.

Indeed

$$\varphi(\psi(\lambda_s(1)))$$

$$= \varphi(g_s(1))$$

$$= \varphi(s)$$

$$= \lambda_s(1).$$

□

Let S be a set and $(A_s)_{s \in S}$ a family of modules.

- Define $\coprod_{s \in S}^{A} A_s$.

- If $A_s = \mathbb{Z}$ for every $s \in S$, find a canonical isomorphism of modules

$$\mathbb{Z}^{(S)} \longrightarrow \coprod_{s \in S}^{A} A_s$$